Epic 3: Data Pre-processing

Story 1: Handling Missing Values

Handling missing values is one of the most important steps in data preprocessing because incomplete data can negatively affect the accuracy and reliability of machine learning models. Missing values occur when some observations are unavailable or not recorded, leading to incorrect analysis and poor prediction performance.

To identify missing values in the flood prediction dataset, the Pandas functions isnull().sum() and isnull().any() are used.

- **isnull().sum() – Calculates the total number of missing values present in each column of the dataset.**
- **isnull().any() – Checks whether any column contains missing values by returning True or False.**

After detecting missing values, appropriate preprocessing techniques such as removing incomplete records or replacing missing values with suitable statistical measures (mean, median, or mode) can be applied to improve data quality.

Outcome

- **Examined the dataset for missing values.**
- **Identified columns containing null values.**
- **Verified the completeness of the dataset before model training.**
- **Improved data quality for accurate flood prediction.**

Figure 7: Detecting missing values using isnull().sum() and isnull().any() in the flood prediction dataset.

```python
#checking null values
dataset.isnull().any()
```

```
Temp           False
Humidity       False
Cloud Cover    False
ANNUAL         False
Jan-Feb        False
Mar-May        False
Jun-Sep        False
Oct-Dec        False
avgjune        False
sub            False
flood          False
dtype: bool
```